

Billel Guelmame

Curriculum Vitae

New York University Abu Dhabi

Abu Dhabi, UAE

+33(0)659937763

+971(0)529652520

billel.guelmame@nyu.edu

<https://guelmame.github.io/>



PhD of Mathematics, University Côte d'Azur, 2020

Personal information

Birth Batna (Algeria), October 2, 1993

Languages Arabic (native), French and English (fluent)

Citizenship Algerian-French

Research interests

Nonlinear hyperbolic partial differential equations

Conservation laws

Nonlinear water waves

Stochastic PDEs

Academic positions

2024–present *Postdoctoral associate* with Taoufik Hmidi at New York University Abu Dhabi, UAE

2022–2024 *Postdoctoral associate* with Julien Vovelle at École Normale Supérieure de Lyon, France

2020–2022 *Temporarily Attached to Education and Research (ATER)*, University Côte d'Azur, France

Education

2017–2020 **PhD** of mathematics at the University Côte d'Azur, Nice, France

Supervisors: Didier Clamond and Stéphane Junca

PhD thesis: On a Hamiltonian regularisation and regularity of entropy solutions of some nonlinear hyperbolic equations

PhD thesis reviewers:

- Adrian Constantin, Professor, University of Vienna
- Philip G. LeFloch, Research Director, Sorbonne University

PhD thesis jury:

- Didier Clamond, Professor, University Côte d'Azur
- Adrian Constantin, Professor, University of Vienna
- Isabelle Gallagher, Professor, École Normale Supérieure Paris
- Sergey Gavriluk, Professor, Aix-Marseille University
- Paola Goatin, Research Director, Inria Sophia Antipolis
- Boris Haspot, Maître de conférences, Paris-Dauphine University
- Stéphane Junca, Maître de conférences, University Côte d'Azur

2016–2017 **Master 2** of Pure and Applied Mathematics at the University Côte d'Azur, Nice, France
Master excellence scholarship, J.A. Dieudonné mathematics laboratory, Nice, France

2015–2016 **Master 1** of mathematics, PDEs and applications at the University of Batna 2, Algeria

2012–2015 **BS degree (Licence)** in fundamental mathematics at the University of Batna 2, Algeria

Teaching experience

- 2025–2026 New York University Abu Dhabi, UAE
◦ *Analysis 1*
- 2024–2025 New York University Abu Dhabi, UAE
◦ *Analysis 1*
- 2023–2024 Claude Bernard University, Lyon, France
◦ *Maths - Coursus prépa* (Analysis and Algebra)
- 2021–2022 University Côte d’Azur, Nice, France
◦ *Maths 2* (Linear Algebra)
◦ *Maths Approfondissements 1* (Arithmetic)
◦ *Maths Approfondissements 2* (Calculus)
◦ *Maths enjeux 1* (Introductory mathematics for first-year students)
◦ *Compléments de Mathématiques* (Complex Numbers)
- 2021–2022 INSPE, Nice, France
◦ *Preparation Course for Master’s in “Métiers de l’Enseignement, de l’Education et de la Formation” (MEEF)*
- 2020–2021 University Côte d’Azur, Nice, France
◦ *Maths Fondements 2* (Calculus)
◦ *Mathematical Methods: Mathematics and Engineering* (Basic mathematics for physics students: derivatives, integrals, curves, series, Fourier transformations)
- 2019–2020 University Côte d’Azur, Nice, France
◦ *Mathematical Methods: Mathematics and Engineering* (Basic mathematics for physics students: derivatives, integrals, curves, series, Fourier transformations)
- 2018–2019 University Côte d’Azur, Nice, France
◦ *Maths Fondements 2* (Calculus)

Conferences and workshops

- May 2025 Theoretical and numerical analysis of multi-scale equations, Lyon, France
- Dec 2024 The 14th AIMS Conference, Abu Dhabi, UAE
- Nov 2023 Journées EDP Rhône-Alpes-Auvergne, Clermont-Ferrand, France
- June 2023 Summer school “New Trends in Mathematical Fluid Dynamics”, Grenoble, France
- March 2023 Journées Jeunes EDPistes, Tours, France
- March 2023 Local and non-local aspects in fluid mechanics, Lyon, France
- Dec 2022 Conference MathFlows, CIRM, Marseille, France
- Nov 2022 Journées EDP Rhône-Alpes-Auvergne, Lyon, France
- June 2022 Summer school on fluids and turbulence, Lyon, France
- June 2022 International Conference on Hyperbolic Problems, Theory, Numerics and Applications, Málaga, Spain
- June 2022 Journées Nice–Toulon–Marseille, Porquerolles, France
- Nov 2021 About the Saint-Venant equations, University Aix-Marseille, France
- Aug 2021 Advanced Summer School in Mathematical Fluid Dynamics, Cargese, Corsica, France
- Jun 2021 Nice–Toulon–Marseille days, Porquerolles, France

Sep 2019 Inhomogeneous Flows: Asymptotic Models and Interface Evolution, CIRM, Marseille, France
 Jun 2019 Waves Côte d’Azur, Nice, France
 May 2019 Colloquium of LJAD’s PhD students, Barcelonnette, France
 Nov 2018 GDR MecaWave, Fréjus, France
 Jun 2018 PDE days 2018, Obernai, France
 May 2018 Nice–Toulon–Marseille days, Porquerolles, France
 May 2018 Colloquium of LJAD’s PhD students, Peyresq, France

Selected talks

June 2026 Séminaire EDP, Université Marie & Louis Pasteur, Besançon, France
 April 2026 Mathematics Seminar at Koç University, Turkey
 May 2025 Theoretical and numerical analysis of multi-scale equations, UMPA, Lyon, France
 Feb 2025 Math Postdoc Seminar, NYUAD, Abu Dhabi, UAE
 Dec 2024 The 14th AIMS Conference, section *Recent Developments in Nonlinear and Nonlocal Evolution Equations*, Abu Dhabi, UAE
 Dec 2024 The 14th AIMS Conference, section *Dynamics and Numerics of Stochastic Differential Equations*, Abu Dhabi, UAE
 Oct 2024 Seminar Analyse, IMA, Strasbourg, France
 Mar 2024 Seminar A3, Amiens, France
 Feb 2024 *Virtual*. PM-EDP Seminar, Galilée Institute, France
 Feb 2024 MAC Seminar at IMT, Toulouse, France
 Jan 2024 Applied Math And Analysis Seminar at Duke University, United States
 Nov 2023 EDP Rhône-Alpes-Auvergne 2023 Days, Clermont-Ferrand, France
 Oct 2023 UMPA, ENS Lyon, France
 Mar 2023 Young EDPists Days, Tours, France
 Dec 2022 *Virtual*. Quebec Analysis and Related Fields Seminar, Laval University, Canada
 Dec 2022 MathFlows Conference, CIRM, Marseille, France
 Nov 2022 “Jeunes analystes et modélisateurs lyonnais” Days, Lyon, France
 Oct 2022 Seminar of the *Physique Mathématique - EDP* team, Bordeaux, France
 Jun 2022 International Conference on Hyperbolic Problems, Málaga, Spain
 Jun 2022 Seminar of the EDPAN team, LJAD, Nice, France
 Oct 2021 *Virtual*. Tassili Sunday Maths Seminar
 Jan 2021 *Virtual*. TIFR, Bangalore, India
 Nov 2018 MecaWave, Fréjus, France
 Mar 2018 LJAD, Nice, France

Research visits

June 2026 Université Marie & Louis Pasteur, Besançon, France (10 days). Worked with Zied Ammari on statistical solutions of the Navier–Stokes equations
 April 2026 Koç University, Istanbul, Turkey (one week). Worked with Haroune Houamed on the ill-posedness of the Hunter–Saxton equation in critical spaces.
 Feb–Mar 2019 TIFR–CAM, Bangalore, India (two weeks). Worked with Stéphane Junca, Shyam Ghoshal, and Animesh Jana on the regularizing effect of entropy solutions for certain nonlinear hyperbolic equations and their optimality.

Research projects

- 2020–2022 IFCAM Project: Conservation Laws: BV^s , Control and Interfaces
◦ Principal Investigators: S.S. Ghoshal (TIFR, Bangalore), S. Junca (UCA, Nice)

Service and organization

- 2026–present Co-organizer of the [Tassili Sunday Maths Seminar \(TSMS\)](#)
2024–present Co-organizer of the [Math Postdoc Seminar](#) at NYUAD
2019 Co-organizer of the Colloquium of LJAD’s PhD students, Barcelonnette, France

Publications

• Preprints

16. B. Guelmame. On the singular limit of the Camassa–Holm equation. 2026.
[arXiv:2609.08227](#)
15. B. Guelmame and H. Houamed. Strong ill-posedness in critical and subcritical regimes for the Hunter–Saxton equation. 2026.
[arXiv:2609.02523](#)
14. P. Alphonse, B. Guelmame and J. Vovelle. A quantitative averaging lemma for spatially dependent vector fields. 2026.
[arXiv:2604.15884](#) [hal-05597049](#)
13. B. Guelmame, T. Hmidi, H. Houamed and F. Rousset. Singular traveling waves for the Euler–Poisson system. 2026.
[arXiv:2604.14997](#) [hal-05602127](#)

• Published/Accepted papers

12. B. Guelmame and J. Vovelle. A Smoluchowski–Kramers approximation for the stochastic variational wave equation. 2026.
[arXiv:2511.13567](#) [hal-05369528](#) [J. Dynam. Differential Equations.](#)
11. B. Guelmame and J. Vovelle. Global dissipative martingale solutions to the variational wave equation with stochastic forcing. 2024.
[arXiv:2403.03034](#) [hal-04282928](#) [Stochastics and Partial Differential Equations.](#)
10. B. Guelmame and H. Houamed. Convergence rate for a regularized scalar conservation law. 2024.
[arXiv:2403.03794](#) [hal-04493491](#) [Zeitschrift für Angewandte Mathematik und Physik.](#)
9. B. Guelmame. On a Hamiltonian regularization of scalar conservation laws. 2024.
[arXiv:2403.02218](#) [hal-02512810](#) [Discrete Contin. Dyn. Syst. A.](#)
8. B. Guelmame, S. Junca, D. Clamond and R. Pego. Global weak solutions of a Hamiltonian regularised Burgers equation. 2024.
[arXiv:2402.15545](#) [hal-02478872](#) [J. Dynam. Differential Equations.](#)
7. B. Guelmame. Global weak solutions of the Serre–Green–Naghdi equations with surface tension. 2023.
[arXiv:2403.02214](#) [hal-03585433](#) [Annales de l’Institut Henri Poincaré C, Analyse non linéaire.](#)
6. B. Guelmame. On the blow-up scenario for some modified Serre–Green–Naghdi equations. 2022.
[arXiv:2403.02208](#) [hal-03264688](#) [Nonlinear Analysis.](#)
5. B. Guelmame, D. Clamond and S. Junca. Local well-posedness of a Hamiltonian regularisation of the Saint-Venant system with uneven bottom. 2022.
[arXiv:2402.15544](#) [hal-02915262](#) [Methods and Applications of Analysis.](#)
4. C. Bourdarias, A.P. Choudhury, B. Guelmame and S. Junca. Entropy solutions in BV^s for a class of triangular systems involving a transport equation. 2022.
[arXiv:2402.16820](#) [hal-02895603](#) [SIAM J. Math. Anal.](#)

3. B. Guelmame, D. Clamond and S. Junca. Hamiltonian regularisation of the unidimensional barotropic Euler equations. 2022.
[arXiv:2402.15261](#) [hal-02907304](#) [Nonlinear Anal. Real World Appl.](#)
2. S. Ghoshal, B. Guelmame, A. Jana and S. Junca. Optimal regularity for all time for entropy solutions of conservation laws in BV^s . 2020.
[arXiv:2402.15250](#) [hal-02495036](#) [NoDEA Nonlinear Differential Equations Appl.](#)
1. B. Guelmame, S. Junca and D. Clamond. Regularizing effect for conservation laws with a Lipschitz convex flux. 2019.
[arXiv:2402.14967](#) [hal-01943834](#) [Commun. Math. Sci.](#)